

Quiz 1: MA 5104, Department of Mathematics, IIT Madras

Name : Statistical Inference and Learning Maximum Marks: 100

Roll no: ID255027 Name: PUNEET A

- The exam duration is 50 minutes.
- Write **ALL Formulas/Theorems/Results** used.
- No need to write the entire question on your answer sheets.
- Write your roll number and name on both the question paper and the answer sheet.
- Calculators are allowed, but sharing them is NOT allowed.
- You do not need to return the question paper. Only the answer sheet will be collected.
- If the answer is a rational number, leave it as a fraction or in decimals up to 3 decimal places.

1. (10 marks) Circle ONE correct answer for each of the sub-parts:

(a) Which of the following statements about the confidence interval (based on the same sample) is/are correct?

CI ↑, width ↑

- ☒ I. A 95% confidence interval is always wider than a 90% confidence interval.
- II. A 95% confidence interval is always shorter than a 90% confidence interval. ✗
- ☒ III. A 95% confidence interval is always shorter than a 99% confidence interval.
- IV. A 95% confidence interval is always wider than a 99% confidence interval.

☒ (A) I and III

(B) II and IV

- (C) I and II
 - (D) II and III
 - (E) None of the above
- (b) What is the main reason that $\bar{X}$ becomes a more reliable estimator of μ as n increases?
- ☒ (A) Its expectation approaches μ
 - (B) Its bias becomes negative
 - (C) Its variance decreases
 - (D) The population variance decreases
- ☒ (c) The term "sampling distribution" of a statistic refers to:
- (A) The distribution of the population from which the sample was drawn
 - (B) The distribution of the observed data values
 - (C) The distribution of the unknown parameter
 - ☒ (D) The distribution of the statistic over repeated random samples
- (d) Suppose $x_1, \dots, x_n$ are the observed values from a random sample, and let $L(\theta | x_1, \dots, x_n)$ denote the likelihood function. The maximum likelihood estimator of θ is the value that:
- ☒ (A) Maximizes the probability of the parameter θ
 - (B) Makes the observed data most likely among all possible values of θ
 - (C) Minimizes the variance of the estimator
 - (D) Makes the estimator unbiased

2. (10 marks) True/False question. Circle T for true and F for false.

- (a) T ~~F~~ The bias of an estimator $\hat{\theta}$ cannot be negative.
- (b) T ~~F~~ The MSE of an estimator $\hat{\theta}$ can be negative.
- (c) T ~~F~~ An estimator $\hat{\theta}$ is unbiased if $E[\hat{\theta}] \geq 0$.
- (d) T ~~F~~ An estimator is a random variable having a distribution
- (e) T ~~F~~ A good estimator should have high bias and low variance.

3. To determine the proportion of people in a certain population with a certain trait, a random sample from the population will be taken to estimate that proportion p . Note that $z_{0.025} = 1.96$. Find a sample size n necessary to estimate p to within 0.05 with probability 0.95 in the following two cases

- (a) (5 marks) The proportion p is thought to be 0.2. 246
- (b) (5 marks) No information about p is available. 385

4. Let X and Y be independent, X having a normal distribution with mean 2θ and variance 2 and Y having a normal distribution with mean θ and variance 7.

(1) (10 marks) Find the MSE of the estimator $\hat{\theta}_1 = (X + Y)/3$. 1

(2) (10 marks) Find the MSE of the estimator $\hat{\theta}_2 = (4X - Y)/7$. 0.51020

(3) (5 marks) Which estimator is better? Provide reasoning

5. The following data represent the length of time, in days, to recovery for patients randomly treated with one of two medications to clear up severe bladder infections:

$$\text{Medication 1 } n_1 = 14 \quad \bar{x}_1 = 17 \quad s_1^2 = 1.5$$

$$\text{Medication 2 } n_2 = 16 \quad \bar{x}_2 = 19 \quad s_2^2 = 1.8$$

(a) (10 marks) Find a 99% confidence interval for the difference $\mu_1 - \mu_2$, in the mean recovery times for the two medications, assuming normal populations with equal variances. Note that $t_{0.005, 28} = 2.763$. $-3.30303, -0.69697$

(b) (5 marks) Interpret the CI in (a). Draw a conclusion from the CI as to which of medicine 1 or 2 should be adopted if the goal is to reduce the length of time to recovery.

(c) (10 marks) Note that $F_{0.05, 13, 15} = 2.45$ and $F_{0.05, 15, 13} = 2.53$. Construct a 90% confidence interval for σ_1^2 / σ_2^2 in part (a). $(0.34012, 2.108249)$

(d) (5 marks) Were we justified in assuming that $\sigma_1^2 = \sigma_2^2$ when we constructed the confidence interval for $\mu_1 - \mu_2$?

6. Let $X_1, \dots, X_n$ be a random sample from a Poisson distribution with probability mass function

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}, \quad x = 0, 1, 2, \dots, \quad \lambda > 0.$$

You may use the fact that $E(X) = \lambda$.

(a) (5 marks) Find the method-of-moments estimator of λ .

$\bar{X}$

(b) (10 marks) Find the maximum likelihood estimator of λ .

$\bar{X}$